

6 The fastest recorded speed by a human was set by Usain Bolt sprinting in 2009. He achieved a speed of 12.3 metres per second.

- (a) (i) Name the main type of muscle fibre that is being used in the leg muscles of a sprinter.

(1)

- (ii) During a sprint race, leg muscles produce lactate.

Describe what happens to the lactate at the end of the race.

(3)

(iii) Which of the following statements are features of muscle fibres in a long-distance runner?

(1)

1. have reduced blood capillary network
2. have larger number of mitochondria
3. contain a low concentration of myoglobin

- ☐ **A** 1 only
- ☐ **B** 2 only
- ☐ **C** 1 and 3 only
- ☐ **D** 2 and 3 only

(iv) Which of the following explains how muscles contract?

(1)

- ☐ **A** Fick's law
- ☐ **B** fluid mosaic model
- ☐ **C** lock and key hypothesis
- ☐ **D** sliding filament model

*(b) Exercise has many benefits for the health and wellbeing of an individual.

An investigation was carried out on non-athletes and athletes.

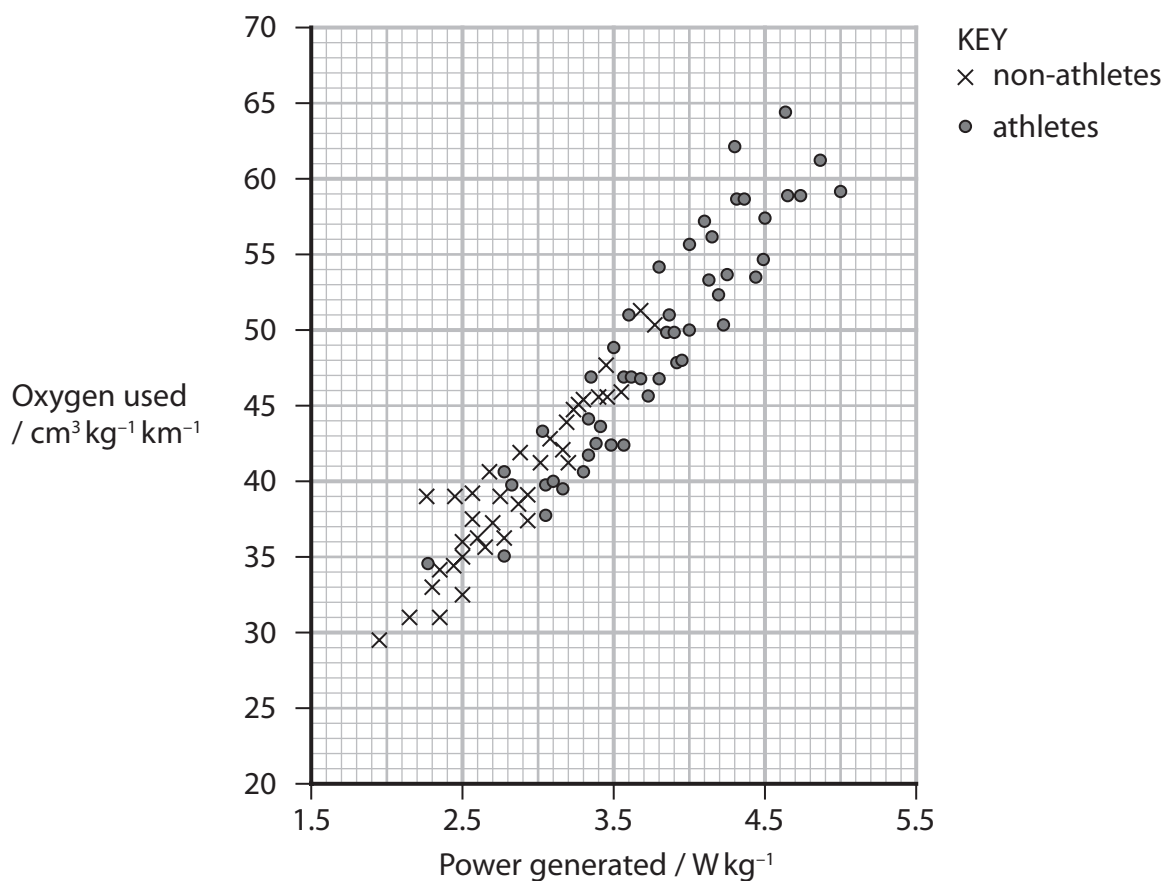
All participants did a series of physical exercises and several measurements were taken.

The means of the measurements are given in the table.

Mean measurement	Non-athletes (n=11)	Athletes (n=10)
Resting heart rate / bpm	71	66
Peak heart rate / bpm	196	190
Maximum volume of oxygen used under anaerobic conditions / $\text{cm}^3 \text{kg}^{-1} \text{minute}^{-1}$	20	35
Maximum volume of oxygen used under aerobic conditions / $\text{cm}^3 \text{kg}^{-1} \text{minute}^{-1}$	39	57

In a different investigation the relationship between power generated and oxygen used was studied in athletes and non-athletes.

The results are shown in the graph.



Discuss the differences in the results for the athletes and non-athletes in these two investigations.

Use your knowledge and information in the table and graph to support and explain your answer.

(6)

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(Total for Question 6 = 12 marks)